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Co5 – AT3
MLA 0201 – Fundamentals of Machine Learning

Annexure A

Mock Interview

- Q1. “Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?” (5 Marks – 8 Mins)
- Q2. “Imagine you are asked in an interview to design an image denoising solution. How would Markov Random Fields help model neighborhood pixel dependencies?” (5 Marks – 8 Mins)
- Q3. “If a company wants to build a speech recognition system, how would you apply a Hidden Markov Model to process sequential audio signals?” (5 Marks – 8 Mins)
- Q4. “During a technical interview, how would you explain the use of Conditional Random Fields for sequence labeling tasks such as part-of-speech tagging?” (5 Marks – 8 Mins)
- Q5. “Suppose an interviewer asks how conditional independence simplifies probabilistic inference. How would you justify it with a graphical model example?” (5 Marks – 8 Mins)
- Q6. “In a healthcare diagnosis problem, how would you compare directed graphical models and undirected graphical models for representing symptom relationships?” (5 Marks – 8 Mins)
- Q7. “If patient symptom data is partially missing, how would you perform Bayesian inference to still make an accurate diagnosis?” (5 Marks – 8 Mins)
- Q8. “How would you use a Hidden Markov Model to analyze customer purchase sequences in an e-commerce recommendation system?” (5 Marks – 8 Mins)
- Q9. “In an NLP interview task, how would you explain why Conditional Random Fields often outperform Hidden Markov Models in named entity recognition?” (5 Marks – 8 Mins)
- Q10. “Suppose you are asked to improve fraud detection. How would graphical model learning and generalization enhance prediction accuracy?” (5 Marks – 8 Mins)

1. Bayesian Networks for Predicting Machine Failures

Answer:

I would use a **Bayesian Network (BN)** to represent the probabilistic relationships between different machine conditions and the possibility of failure.

First, I would identify important variables such as **temperature, vibration, pressure, lubricant level, machine age, and machine failure**. These variables would be represented as nodes in the network. Directed edges would represent relationships between the variables.

For example:

Temperature → Machine Failure

Vibration → Machine Failure

Lubricant Level → Temperature → Machine Failure

I would collect historical sensor data and use it to learn the probabilities associated with these relationships. When new sensor readings are received, Bayesian inference can calculate:

$P(\text{Machine Failure} \mid \text{Sensor Observations})$

If this probability becomes higher than a predefined threshold, the system can generate an early warning and schedule preventive maintenance.

The main advantage is that Bayesian Networks can handle **uncertainty and incomplete sensor information**. Therefore, they are useful for industrial predictive maintenance.

Conclusion:

A Bayesian Network would allow the company to combine sensor information and historical knowledge to estimate the probability of machine failure before it actually occurs.

2. Markov Random Fields for Image Denoising

Answer:

A **Markov Random Field (MRF)** is useful for image denoising because neighboring pixels in an image are usually related to each other.

I would represent every pixel as a node in an undirected graphical model. Each pixel would be connected to its neighboring pixels, such as the pixels above, below, left, and right.

For example:

Pixel A $\leftrightarrow$ Pixel B $\leftrightarrow$ Pixel C

The observed noisy image would provide the input evidence, while the MRF would estimate the most likely clean image.

The model would have two main components:

1. **Data term:** Measures how closely the estimated pixel value matches the observed noisy pixel.
2. **Smoothness term:** Encourages neighboring pixels to have similar values unless there is a meaningful edge.

The objective would be to find the pixel configuration with the **lowest overall energy** or highest probability.

For example, if one pixel is very different from all its neighbors because of random noise, the MRF can determine that the pixel is likely corrupted and assign a value consistent with its neighborhood.

Conclusion:

MRFs are effective for image denoising because they explicitly model **spatial dependencies between neighboring pixels**, helping preserve important structures while removing noise.

3. Hidden Markov Model for Speech Recognition

Answer:

I would use a **Hidden Markov Model (HMM)** because speech is sequential data, where the current sound is related to previous sounds.

In an HMM, the actual speech units, such as **phonemes or words**, can be treated as hidden states. The observed data would be acoustic features extracted from the audio signal.

The process would be:

Speech Audio → Feature Extraction → HMM → Recognized Speech

First, I would divide the audio into small time frames. From each frame, I would extract features such as **MFCCs (Mel-Frequency Cepstral Coefficients)**.

The HMM would contain:

- **States:** Phonemes or speech units
- **Transition probabilities:** Probability of moving from one speech state to another
- **Emission probabilities:** Probability of observing an acoustic feature from a particular state
- **Initial probabilities:** Probability of starting in a particular state

Given the sequence of acoustic observations, algorithms such as the **Viterbi algorithm** can find the most probable sequence of hidden states.

Conclusion:

HMMs are suitable for speech recognition because they model the **temporal and sequential nature of speech** and can estimate the most likely spoken sequence from acoustic observations.

4. Conditional Random Fields for POS Tagging

Answer:

A **Conditional Random Field (CRF)** is a discriminative probabilistic model that is widely used for sequence labeling tasks such as **Part-of-Speech (POS) tagging**.

In POS tagging, the input is a sequence of words and the output is a sequence of tags.

For example:

The → DET

boy → NOUN

runs → VERB

A CRF predicts the complete sequence of tags based on the entire input sentence rather than predicting each word independently.

It can use features such as:

- Current word
- Previous and next words
- Word suffix and prefix
- Capitalization
- Previous POS tag
- Word shape

For example, if the word is "running", the suffix **"-ing"** can be a useful feature for identifying it as a verb.

CRFs model dependencies between neighboring output labels while considering the observed words.

Conclusion:

CRFs are powerful for POS tagging because they consider **context and relationships between neighboring tags**, producing more consistent sequence-labeling results.

5. Conditional Independence and Probabilistic Inference

Answer:

Conditional independence means that two variables become independent when we know the value of another variable.

Consider a simple Bayesian Network:

Rain → **Wet Grass** ← **Sprinkler**

Here, **Rain** and **Sprinkler** may be independent before observing the grass condition. Once we observe whether the grass is wet, their relationship can change.

Another simple example is:

Disease → **Symptoms**

If the disease is known, individual symptoms may become conditionally independent in a simplified model.

Graphical models simplify inference because they allow us to factorize a large joint probability distribution into smaller probability distributions.

Instead of calculating:

P(A, B, C, D, E)

we can use the graph structure to represent it as smaller conditional probabilities.

For a Bayesian Network:

$P(X_1, X_2, \dots, X_n) = \prod P(X_i \mid \text{Parents}(X_i))$

This reduces the number of probabilities that need to be considered.

Conclusion:

Conditional independence reduces computational complexity by allowing a large probability distribution to be **factorized into smaller components**, making probabilistic inference faster and more manageable.

6. Directed vs Undirected Graphical Models in Healthcare

Answer:

I would compare **directed graphical models**, such as Bayesian Networks, with **undirected graphical models**, such as Markov Random Fields.

Directed Graphical Models

A Bayesian Network uses directed edges to represent probabilistic dependencies.

For example:

Disease → Fever

Disease → Cough

Disease → Fatigue

It is useful when we want to represent **causal or directional relationships** and perform diagnostic reasoning.

Undirected Graphical Models

An MRF uses undirected edges.

For example:

Fever — Cough — Fatigue

The connections represent relationships between variables without specifying a causal direction.

MRFs are useful when relationships are **symmetric**, such as correlations between symptoms or neighboring medical measurements.

Comparison

Feature	Bayesian Network	Markov Random Field
Graph	Directed	Undirected
Direction	Yes	No
Causal interpretation	Possible	Not directly
Example	Disease → Symptom	Symptom ↔ Symptom
Useful for	Diagnosis and causal reasoning	Dependency modeling

Conclusion:

For a healthcare diagnosis system where causal relationships are important, I would generally prefer a **Bayesian Network**. If the goal is to model mutual dependencies between variables, an **MRF** may be more suitable.

7. Bayesian Inference with Missing Patient Symptoms

Answer:

If some patient symptoms are missing, I would use **Bayesian inference** because Bayesian Networks can reason under uncertainty and incomplete observations.

Suppose we have:

Disease → Fever

Disease → Cough

Disease → Fatigue

Now assume the patient's cough information is missing, but fever and fatigue are available. Instead of treating the missing symptom as a negative symptom, I would simply treat it as **unobserved**.

The system can calculate:

$P(\text{Disease} \mid \text{Observed Symptoms})$

Using Bayes' theorem:

$P(\text{Disease} \mid \text{Evidence}) \propto P(\text{Evidence} \mid \text{Disease}) \times P(\text{Disease})$

The inference engine considers all possible values of the missing symptom and sums or marginalizes over them.

For example, if fever and fatigue strongly indicate a disease, the system can still produce a high probability even though cough information is unavailable.

Conclusion:

Bayesian inference is useful for incomplete medical data because it does not require every symptom to be present. It uses the **available evidence and prior probabilities** to estimate the most likely diagnosis.

8. HMM for Customer Purchase Sequences

Answer:

I would use an **HMM** to model customer purchasing behavior because customers generate sequences of actions over time.

The hidden states could represent customer interests or shopping intentions, such as:

- **Browsing**
- **Interested**
- **Ready to Purchase**
- **Repeat Customer**

The observed events could be:

- Product viewed
- Product searched
- Product added to cart
- Product purchased

For example:

Browsing → Interested → Ready to Purchase → Purchase

The transition probabilities represent how likely a customer is to move from one state to another.

The emission probabilities represent how likely a particular customer action is in each hidden state.

After learning the HMM from historical customer sequences, the system can estimate the customer's current hidden state and predict the next likely action.

For example, if a customer repeatedly views laptops and adds accessories to the cart, the model may identify a high probability of being in a **Ready-to-Purchase** state and recommend relevant laptop products.

Conclusion:

HMMs can help recommendation systems understand **customer behavior over time** and predict future interests from sequential purchase or browsing patterns.

9. Why CRFs Often Outperform HMMs in Named Entity Recognition

Answer:

Both **HMMs and CRFs** can be used for sequence labeling, but CRFs often perform better in Named Entity Recognition (NER) because CRFs can use many overlapping and dependent features from the input.

An HMM is a **generative model**. It models:

P(Observations, States)

It generally relies on assumptions about how observations are generated from hidden states.

A CRF is a **discriminative model**. It directly models:

P(Labels | Observations)

For NER, useful features include:

- Current word
- Previous and next words
- Capitalization
- Prefixes and suffixes
- Word shape
- Surrounding words
- Previous labels

For example:

"Dr. Ramesh works at Microsoft."

The capitalization and surrounding words can help identify **Ramesh** as a person's name and **Microsoft** as an organization.

CRFs can combine these features while modeling dependencies between neighboring labels.

Conclusion:

CRFs often outperform HMMs in NER because they can incorporate **rich, overlapping features and label dependencies** without requiring the same strong independence assumptions as HMMs.

10. Graphical Model Learning and Generalization for Fraud Detection

Answer:

I would use graphical models to represent relationships among **customers, transactions, devices, locations, merchants, and accounts**.

For example:

Customer $\rightarrow$ Transaction $\leftarrow$ Merchant

and:

Customer $\rightarrow$ Device

Transaction $\rightarrow$ Location

Fraudulent transactions often have relationships that are difficult to detect using isolated transaction features.

First, I would collect historical transaction data and learn the parameters of the graphical model. The model would estimate the probability that a transaction is fraudulent based on the observed evidence.

For example:

P(Fraud | Transaction Features, Customer Behavior, Device, Location)

Graphical model learning can discover important dependencies between variables.

Generalization is important because the model must work not only on previously seen transactions but also on **new and unseen transactions**.

To improve generalization, I would:

1. Use representative training data.
2. Avoid overfitting.
3. Validate the model on unseen data.
4. Use appropriate regularization.

For example, if the same device is suddenly associated with many accounts and unusual transactions occur from different locations, the graphical model can combine these relationships and increase the estimated fraud probability.

Conclusion:

Graphical model learning helps discover complex relationships in transaction data, while good generalization ensures that these learned patterns remain useful for new and previously unseen fraud cases.